COMMA SEPARATED VALUE

CSV FILE

- **CSV** is a simple **file format** used to store tabular data, such as a spreadsheet or database.
- **CSV** stands for "comma-separated values".
- A comma-separated values file is a delimited text file that uses a comma to separate values.
- Each line of the file is a data record. Each record consists of one or more fields, separated by commas. The use of the comma as a field separator is the source of the name for this file format

CSV file handling in Python

- To perform read and write operation with CSV file, we must import **csv** module.
- open() function is used to open file, and return file object.

Reading from CSV file

- import csv module
- Use open() to open csv file, it will return file object.
- Pass this file object to reader object.
- Perform operation you want

Example : Reading from CSV File

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

```
import csv
with open('myfile.csv') as csvfile:
        myreader = csv.reader(csvfile,delimiter=',')
        print("%10s"%"EMPNO","%20s"%"EMP NAME","%10s"%"SALARY")
        print("=================================================")
        for row in myreader:
                print("%10s"%row[0],"%20s"%row[1],"%10s"%row[2])
```

```
     EMPNO                EMP NAME      SALARY
=================================================
         1                    Amit        6000
         2           Suresh Kumar         8000
         3                  Gabbar       75000
```

OUTPUT

GO TO NEXT SLIDE TO KNOW HOW TO CREATE CSV FILE

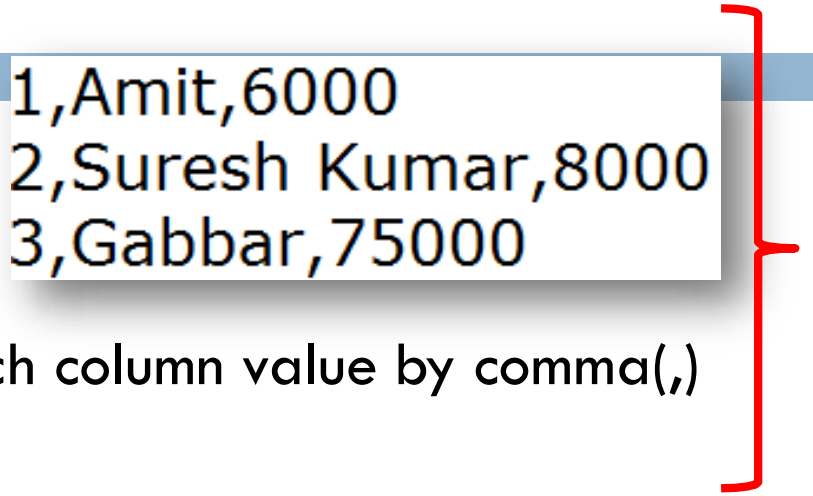
How to create CSV file

- Method 1 (From MS-Excel):
  - Open Excel, delete all the sheet except sheet 1
  - Type all the data, in separate cells
  - Save it as **CSV** file in your desired location.
  - If any warning comes, click on „YES"
  - When you close the excel, choose „NO"
  - Now file is created at your desired location, go and double click or open with notepad to check the content

A	B	C
1	Amit	6000
2	Suresh Kumar	8000
3	Gabbar	75000

How to create CSV file

- Method 2 (From Notepad):
  - Open Notepad
  - Type record by separating each column value by comma(,)
  - Every record in separate line
  - Save it by giving extension .csv (PUT THE NAME IN DOUBLE QUOTES TO ENSURE .TXT WILL NOT BE APPENDED WITH FILE NAME FOR E.G. if you want it to save with name emp then give name as “emp.csv” in double quotes
  - File is created close it and double click to open and check



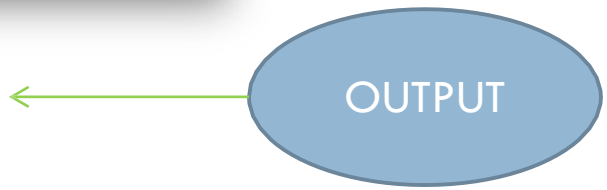
Example : Counting number of records

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

```
import csv
with open('myfile.csv') as csvfile:
        myreader = csv.reader(csvfile,delimiter=',')
        count=0
        print("%10s"%"EMPNO","%20s"%"EMP NAME","%10s"%"SALARY")
        print("================================================")
        for row in myreader:
                print("%10s"%row[0],"%20s"%row[1],"%10s"%row[2])
                count+=1
        print("================================================")
        print("%30s"%"TOTAL RECORDS :",count)
        print("================================================")
```

```
     EMPNO                 EMP NAME      SALARY
================================================
         1                     Amit        6000
         2             Suresh Kumar        8000
         3                   Gabbar       75000
================================================
                TOTAL RECORDS : 3
================================================
```



Example : Sum of Salary and counting employee getting more than 70000

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

```
import csv
with open('myfile.csv') as csvfile:
        myreader = csv.reader(csvfile,delimiter=',')
        count=0
        sum=0
        print("%10s"%"EMPNO","%20s"%"EMP NAME","%10s"%"SALARY")
        print("================================================")
        for row in myreader:
                print("%10s"%row[0],"%20s"%row[1],"%10s"%row[2])
                sum+=int(row[2])
                if int(row[2])>70000:
                        count+=1
        print("================================================")
        print("%30s"%"SUM OF SALARY :",sum)
        print("%40s"%"#EMPLOYEE GETTING SALARY >70000 :",count)
        print("================================================")
```

Example : Sum of Salary and counting employee getting more than 70000

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

```
import csv
with open('myfile.csv') as csvfile:
        myreader = csv.reader(csvfile,delimiter=',')
        count=0
        sum=0
        print("%10s"%"EMPNO","%20s"%"EMP NAME","%10s"%"SALARY")
        print("=================================================")
        for row in myreader:
                print("%10s"%row[0],"%20s"%row[1],"%10s"%row[2])
                sum+=int(row[2])
                if int(row[2])>70000:
                        count+=1
        print("=================================================")
        print("%30s"%"SUM OF SALARY :",sum)
        print("%40s"%"#EMPLOYEE GETTING SALARY >70000 :",count)
        print("=================================================")
```

```
     EMPNO                EMP NAME      SALARY
=================================================
         1                    Amit        6000
         2           Suresh Kumar         8000
         3                  Gabbar       75000
=================================================
               SUM OF SALARY : 89000
      #EMPLOYEE GETTING SALARY >70000 : 1
=================================================
```

OUTPUT

Writing data in CSV file

- import csv module
- Use open() to open CSV file by specifying mode “w” or “a”, it will return file object.
- “w” will overwrite previous content
- “a” will add content to the end of previous content.
- Pass the file object to writer object with delimiter.
- Then use writerow() to send data in CSV file

Example : Writing data to CSV file

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

```
import csv
with open('myfile.csv',mode='a') as csvfile:
        mywriter = csv.writer(csvfile,delimiter=',')
        ans='y'
        while ans.lower()=='y':
                eno=int(input("Enter Employee Number "))
                name=input("Enter Employee Name ")
                salary=int(input("Enter Employee Salary :"))
                mywriter.writerow([eno,name,salary])
                print("## Data Saved... ##")
                ans=input("Add More ?")
```

Example : Writing data to CSV file

```
import csv
with open('myfile.csv',mode='a') as csvfile:
        mywriter = csv.writer(csvfile,delimiter=',')
        ans='y'
        while ans.lower()=='y':
                eno=int(input("Enter Employee Number "))
                name=input("Enter Employee Name ")
                salary=int(input("Enter Employee Salary :"))
                mywriter.writerow([eno,name,salary])
                print("## Data Saved... ##")
                ans=input("Add More ?")
```

```
Enter Employee Number 4
Enter Employee Name AMAN
Enter Employee Salary :80000
## Data Saved... ##
Add More ?Y
Enter Employee Number 5
Enter Employee Name JACKY
Enter Employee Salary :60000
## Data Saved... ##
Add More ?N
```

OUTPUT

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
```

myfile.csv

BEFORE EXECUTION

```
1,Amit,6000
2,Suresh Kumar,8000
3,Gabbar,75000
4,AMAN,80000
5,JACKY,60000
```

myfile.csv

AFTER EXECUTION